

AI Ops Assignment 1

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Question 1

S.No.	Category
a.	Entanglement(CACE)
b.	Undeclared Consumers
c.	Configuration and Glue Code Debt

For case (b), the new updates that need to be added can be added to this version itself, but this current version should also be saved like the older packages in python. This makes sure that the new updates do not break the dashboard's functionality.

Question 2

Run Name	Created ↕	Action	Metrics			Parameters		
			Test accuracy	Train accuracy	Train loss	hidden_layer_si	learning_rate_i	batch_size
exp6	19 minutes ago	Train	0.96392857...	0.99014285...	0.04756811...	(64, 32)	0.001	32
exp5	19 minutes ago	Train	0.11428571...	0.11208928...	2.31665235...	(64, 32)	0.1	64
exp4	19 minutes ago	Train	0.33107142...	0.33580357...	1.73280758...	(64,)	0.01	32
exp3	19 minutes ago	Train	0.11428571...	0.11208928...	2.31648639...	(64,)	0.1	64
exp2	20 minutes ago	Train	0.23328571...	0.23271428...	1.95604690...	(32,)	0.01	32
exp1	20 minutes ago	Train	0.10235714...	0.10192857...	2.31138160...	(32,)	0.1	64

- We can't see that much of a difference between the train or test accuracy, so we can't conclude that it overfit in any case I've tried.
- But, if the number of hidden layers was increased, it is possible to reach a stage where the models overfit.
- Overall the best performing architecture was the one with a learning rate = 0.001, batch size = 32 and hidden layers = (64, 32).
- The models with smaller architectures underfit on the data and weren't able to learn enough, so both the test and train accuracies are very low.
- And having a higher learning rate could cause the weights to oscillate instead of converging which can also lead to sub optimal results.
- So, the best performing model was the one that had the best of both worlds in this context, with a smaller learning rate and more hidden layers.
- The architecture, number and size of hidden layers, would be one of the main hyperparameters that have a large effect on the performance.

```

mlflow.log_param("hidden_layer_sizes", hidden_layer_sizes)
mlflow.log_param("learning_rate_init", learning_rate_init)
mlflow.log_param("batch_size", batch_size)

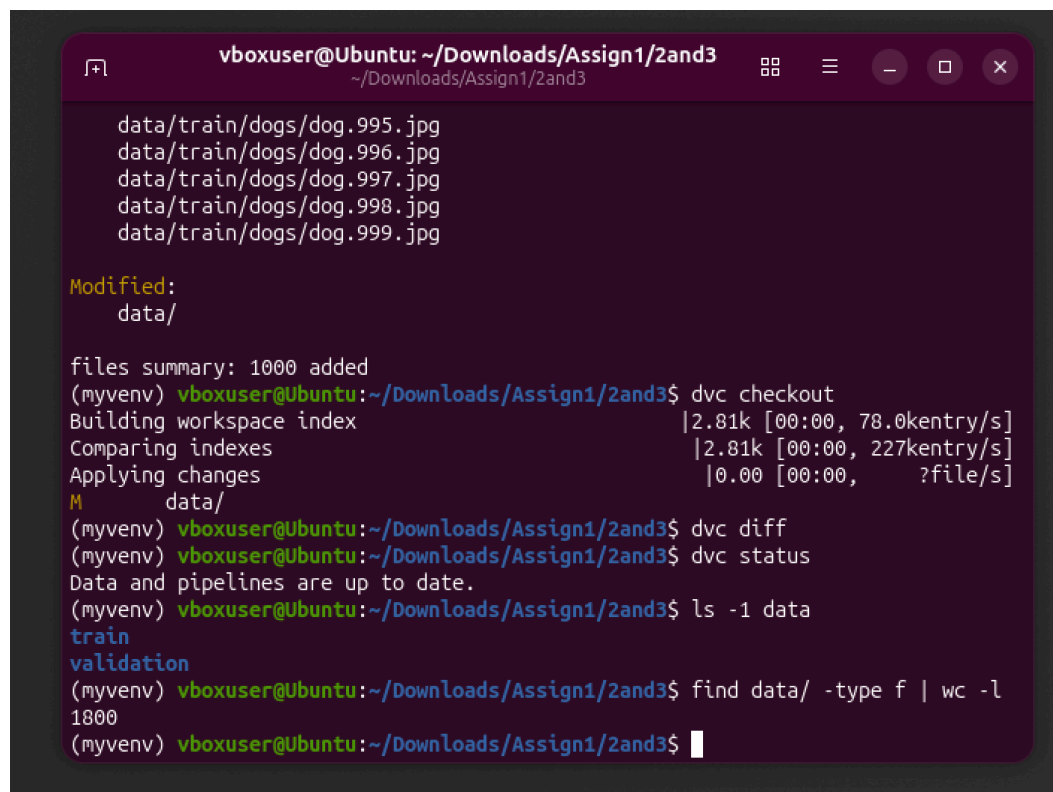
model = MLPClassifier(
    hidden_layer_sizes = hidden_layer_sizes,
    learning_rate_init = learning_rate_init,
    batch_size = batch_size,
    random_state = 42
)

model.fit(X_train, y_train)
train_acc = accuracy_score(y_train, model.predict(X_train))
val_acc = accuracy_score(y_test, model.predict(X_test))
train_loss = model.loss_

mlflow.log_metric("Train accuracy", train_acc)
mlflow.log_metric("Test accuracy", val_acc)
mlflow.log_metric("Train loss", train_loss)

```

Question 3



```

vboxuser@Ubuntu: ~/Downloads/Assign1/2and3
~/Downloads/Assign1/2and3

data/train/dogs/dog.995.jpg
data/train/dogs/dog.996.jpg
data/train/dogs/dog.997.jpg
data/train/dogs/dog.998.jpg
data/train/dogs/dog.999.jpg

Modified:
data/

files summary: 1000 added
(myvenv) vboxuser@Ubuntu:~/Downloads/Assign1/2and3$ dvc checkout
Building workspace index |2.81k [00:00, 78.0kentry/s]
Comparing indexes |2.81k [00:00, 227kentry/s]
Applying changes |0.00 [00:00, ?file/s]
M data/
(myvenv) vboxuser@Ubuntu:~/Downloads/Assign1/2and3$ dvc diff
(myvenv) vboxuser@Ubuntu:~/Downloads/Assign1/2and3$ dvc status
Data and pipelines are up to date.
(myvenv) vboxuser@Ubuntu:~/Downloads/Assign1/2and3$ ls -1 data
train
validation
(myvenv) vboxuser@Ubuntu:~/Downloads/Assign1/2and3$ find data/ -type f | wc -l
1800
(myvenv) vboxuser@Ubuntu:~/Downloads/Assign1/2and3$

```